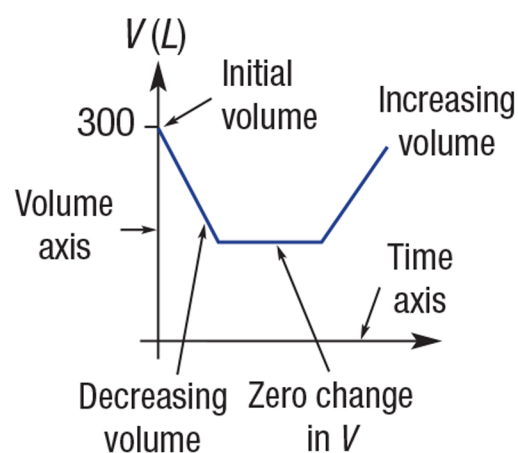
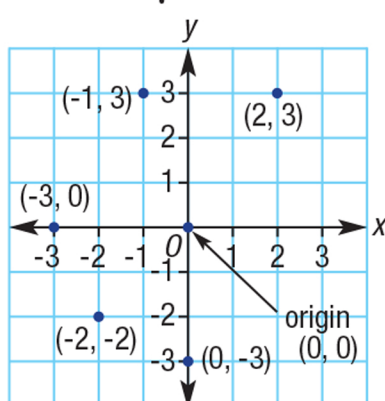


Graphs



Number/Cartesian plane



Rules and tables

$$y = -2x + 3$$

x	-2	-1	0	1	2
y	7	5	3	1	-1

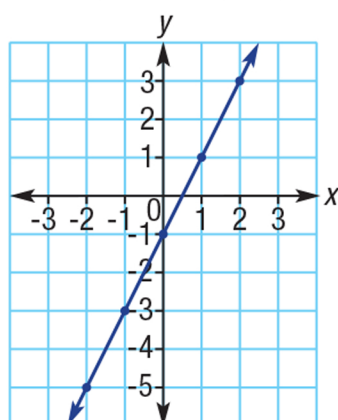
$$\begin{aligned} \text{If } x &= -2 \\ y &= -2 \times (-2) + 3 \\ &= 4 + 3 \\ &= 7 \end{aligned}$$

Plotting straight line graphs

Rule $y = 2x - 1$

Table

x	-2	-1	0	1	2
y	-5	-3	-1	1	3



Straight line graphs

Finding the rule from tables

x	-2	-1	0	1	2
y	-8	-5	-2	1	4

$$\begin{aligned} &\begin{array}{ccccccc} & +3 & +3 & +3 & +3 \\ \downarrow & & & & & \\ y & = \triangle \times x + \square \\ & = 3x + (-2) \\ & = 3x - 2 \end{array} \end{aligned}$$

Applications

- Distance increases by 20 km per hour.

t	0	1	2	3
d	0	20	40	60

$$d = 20t$$

- Volume decreases from 1000 L by 200 L per minute.

t	0	1	2	3	4	5
V	1000	800	600	400	200	0

$$V = -200t + 1000$$

Gradient

